

DATA MINING

ACTIVITY 3 CLASSIFICATION

BY GROUP 1:

HARINI

LING

ADRIANA

Q6: Compare Supervised Learning and Unsupervised Learning with examples.

SUPERVISED LEARNING

- The training data are accompanied by labels indicating the class of observations
- New data is classified based on the training set
- Example : Classification (Predict whether a patient has cancer)

UNSUPERVISED LEARNING

- The class labels of training data is unknown
- Aim of establishing the existence of classes or clusters in the data
- Example : Clustering (Cluster normal data and detect outliers)

Q7: Differentiate between binary classification and multi-class classification. Provide an example for each.

BINARY CLASSIFICATION

- Data being classified into two classes
- Output labels are only two classes (0 or 1, True or False, Yes or No).
- Example : Spam detection - spam vs. not spam

MULTI-CLASS CLASSIFICATION

- Data being classified into more than two classes
- Output labels are more than two classes
- Example : Plant species classification

Source : <https://www.analyticssteps.com/blogs/binary-and-multiclass-classification-machine-learning>

Q8: Discuss common challenges encountered in classification tasks.

1. **Overfitting** - Occurs when the model is too complex and tries to encode more patterns from the training data than it is necessary. The model ends up hyper-focusing only on patterns causing overfitting.
2. **Underfitting** - Occurs when model is too simple or when the data is noisy or contains complex relationships that cannot be modeled by a simple model, leading to poor performance on both training and testing data.
3. **Class imbalance** - Occurs when one class heavily outnumbered the others, causing the models to be biased toward the majority class.

Q9: What is a confusion matrix and how is it used to evaluate classification models? Provide a labeled example.

	Predicted Positive	Predicted Negative	
Actual Positive	TP <i>True Positive</i>	FN <i>False Negative</i>	Sensitivity $\frac{TP}{(TP + FN)}$
Actual Negative	FP <i>False Positive</i>	TN <i>True Negative</i>	Specificity $\frac{TN}{(TN + FP)}$
	Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Q9: What is a confusion matrix and how is it used to evaluate classification models? Provide a labeled example.

Confusion matrix is used to:

- **Solve Imbalanced Data:** Using accuracy as the only metric leads to imbalanced datasets. Thus, using other metrics, such as precision and recall through confusion matrix allows a more balanced view and accurate representation of a dataset.
- **Error Type Differentiator:** Confusion matrix reveals all the different type of errors produced by the model.

Source : https://www.analyticssteps.com/blogs/binary-and-multiclass-classification-machine-learning_

Source : https://www.datacamp.com/tutorial/what-is-a-confusion-matrix-in-machine-learning_

Q10: Explain the concept of overfitting and underfitting in classification models. How can each issue be addressed?

OVERFITTING

- Occurs when the model is too complex and tries to encode more patterns from the training data than it is necessary, causing overfitting.
- Also occurs when a small training set, too small to show the patterns in the data ends up hyper-focusing only on patterns.
- The model is not generalizable, which means that the model won't be very good at predicting the targets of data it has never seen before.
- This issue can be tackled by simplifying the model or using cross-validation.

UNDERFITTING

- A situation where a model has yet to learn the true structure of the data.
- The model is too simple or when the data is noisy or contains complex relationships that cannot be modeled by a simple model
- As a results, the model is unable to fit the training data well, leading to poor performance on both training and testing data, as well as high bias and low variance.
- This issue can be tackled by using more complex model

Q11

Describe the role of cross-validation in evaluating the performance of a classification algorithm.

- **Used to evaluate model performance reliably and avoid overfitting.**
 - **Splits data into k folds → each fold used for testing once, others for training.**
 - **Provides unbiased estimate of generalization error.**
 - **Helps in model selection and hyperparameter tuning.**
 - **More accurate than a simple train/test split, especially for small datasets.**
- documentation**

From Slide: Cross-validation ensures the model is tested on unseen data, improving generalization performance.

Reference: [Tan et al., 2018], Scikit-learn documentation

Q12

How can feature selection improve classification model performance? Mention at least two methods

- **Removes irrelevant/redundant features → improves accuracy and training speed.**
- **Helps prevent overfitting and reduces model complexity.**
- **Makes the model easier to interpret and more efficient.**
- **Two Common Methods:**
 - **Filter Method — uses statistical tests (e.g., chi-square, correlation).**
 - **Wrapper Method — uses model performance (e.g., recursive feature elimination).**

From Slide: Reduces errors and helps classifier focus on useful patterns.

Reference: [Guyon & Elisseeff, 2003]

Q13

Compare decision trees and logistic regression in terms of model interpretability, accuracy, and use cases.

Criteria	Decision Tree	Logistic Regression
Interpretability	Easy to understand (if-else rules)	Coefficients explain feature impact (less intuitive)
Accuracy	Good for nonlinear patterns	Better for linear relationships
Overfitting	Prone if not pruned	Less likely due to linearity
Use Case	Rule-based decisions, business logic	Risk scoring, healthcare, social sciences

From Slide: Decision trees popular for explainability; logistic regression used when linear relationship assumed.

Reference: [Scikit-learn docs], Tim Bock – Analytics Comparisons

Q14

Explain what ensemble methods (e.g., Bagging, Boosting, Random Forest) are and how they improve classification performance.

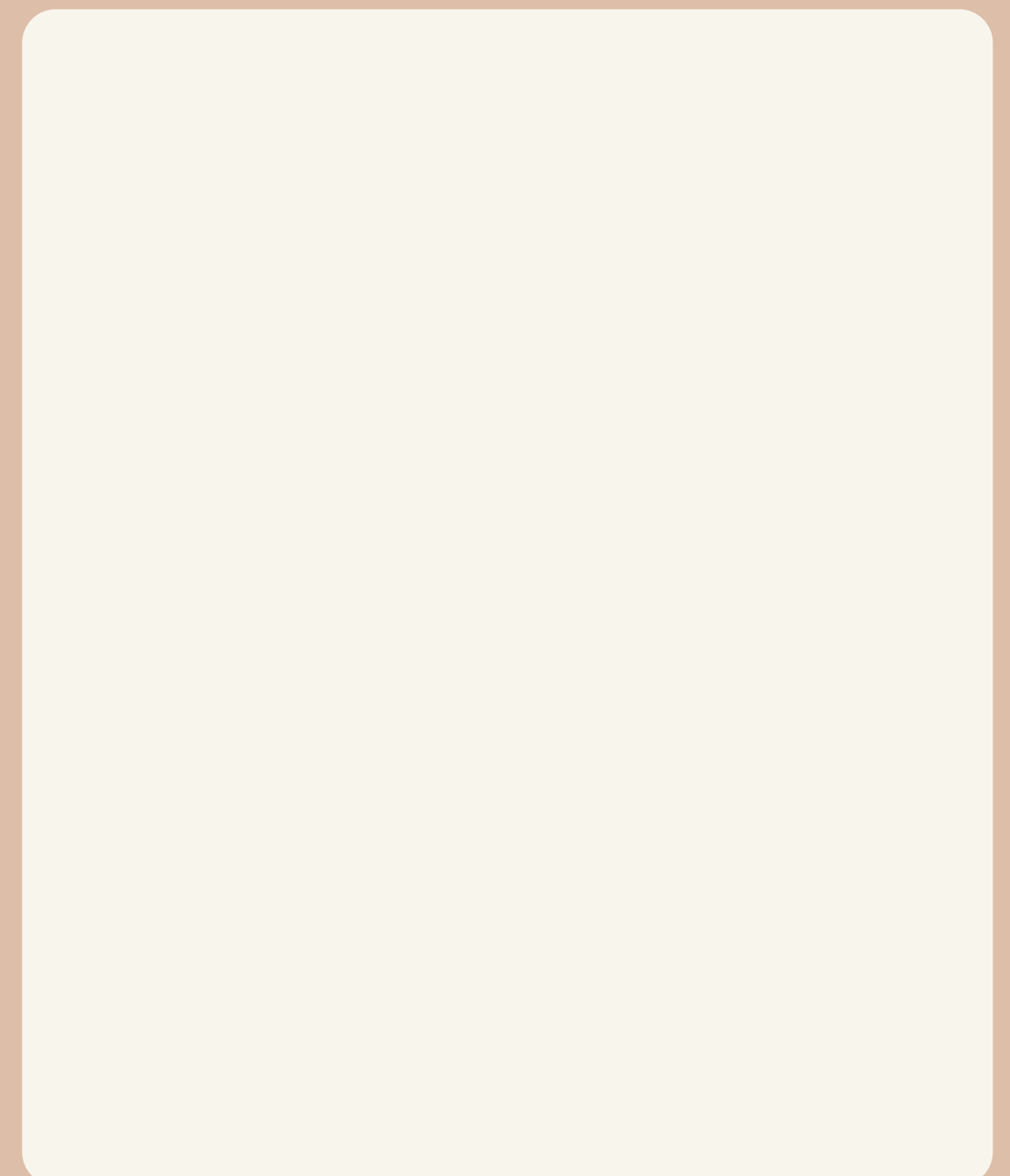
Method	How It Works	Goal	Strengths	Example Algorithm
Bagging	Trains multiple models in parallel on different random bootstrapped subsets	Reduce variance	Prevents overfitting, improves stability	Random Forest
Boosting	Trains models sequentially, each model focuses on previous errors	Reduce bias	Learns from mistakes, improves weak learners	AdaBoost, XGBoost
Random Forest	A form of bagging that also uses random feature selection at each split	Reduce variance, avoid overfitting	Fast, accurate, robust to noise	Random Forest

From Slide: Ensemble methods increase model generalization.
Reference: [Breiman, 1996], Scikit-learn, [Microsoft Docs]

**Today We
Covered:**

A large, empty white rectangular box with rounded corners, intended for taking notes on the topics covered today.

**You Need
To:**

A large, empty white rectangular box with rounded corners, intended for taking notes on the tasks or goals you need to complete.

Homework

Due next lesson:

Chapter 3 questions complete

Read to chapter 5



See you
next lesson!